

View Reviews

Paper ID

228

Paper Title

Short-Term Photovoltaic Forecasting under Data Scarcity using Leave-One-Plant-Out Evaluation

Track Name

Workshop Chileno Sobre Reconocimiento de Patrones

Reviewer #1

Questions

1. Contributions and Strengths: What is the main impact of the work and its relevance to the audience? Evaluate the methodological soundness, the structural coherence, and highlight the main strengths of the paper.

El trabajo aborda la predicción de generación fotovoltaica en condiciones de escasez de datos y cold-start, mediante modelos globales capaces de realizar predicciones sobre plantas no observadas durante el entrenamiento. La temática es pertinente para el CWPR por su aplicación de aprendizaje automático, deep learning, transferencia de conocimiento y análisis de series temporales.

El estudio presenta una evaluación sobre 94 plantas fotovoltaicas chilenas, utilizando modelos TFT, Informer, N-HiTS, LSTM y XGBoost. Una fortaleza metodológica es el protocolo Leave-One-Plant-Out (LOPO), orientado a evaluar la generalización hacia nuevas plantas y controlar el data leakage. También resulta favorable la consideración de horizontes de 24 horas y 7 días, el análisis estacional y la evaluación mediante métricas puntuales y probabilísticas.

La evaluación de incertidumbre mediante cuantiles P5/P50/P95, cobertura de intervalos y pinball loss constituye otro aspecto positivo, permitiendo identificar problemas de calibración en algunos modelos. Los resultados también muestran que XGBoost obtiene un desempeño competitivo frente a arquitecturas más complejas, aportando información relevante respecto de la relación entre complejidad y desempeño.

2. Weaknesses and Suggestions: Identify fixable areas for improvement (e.g., phrasing, lack of detail) as well as major ones (severe structural or methodological flaws). Provide specific suggestions on what to add, remove, or correct for the final version.

La comparación entre modelos globales y locales requiere una interpretación cuidadosa, dado que los primeros operan bajo un escenario zero-shot, mientras que los modelos locales utilizan información histórica de la planta objetivo. Se recomienda precisar mejor el objetivo de esta comparación y diferenciar los análisis realizados bajo condiciones equivalentes.

Aunque LOPO proporciona un mecanismo adecuado para evaluar generalización entre plantas, sería conveniente describir con mayor detalle la separación temporal entre entrenamiento, validación y prueba, así como el procedimiento de selección de hiperparámetros. Los autores reconocen cierto parameter-level leakage debido a la reutilización de hiperparámetros entre folds, aspecto que debería justificarse y discutirse con mayor profundidad.

La comparación de los modelos se basa principalmente en métricas agregadas y medianas entre plantas. Se recomienda incorporar pruebas estadísticas que permitan establecer si las diferencias observadas entre los métodos son significativas, particularmente considerando que el manuscrito plantea el rechazo de una hipótesis de superioridad de los modelos globales.

También sería conveniente incorporar gráficos representativos de las series temporales, mostrando valores observados y predichos para los horizontes de 24 horas y 7 días. En el caso de las predicciones probabilísticas, podrían incluirse P50 y las bandas P5–P95. Esto facilitaría la evaluación cualitativa de la dinámica temporal y de la calibración de la incertidumbre.

Finalmente, se recomienda ampliar el estado del arte reciente, particularmente en estrategias zero-shot/few-shot y métodos de calibración de incertidumbre, con el propósito de contextualizar mejor la contribución respecto de desarrollos recientes.

3. What is the overall score of the submission?

accept

Reviewer #2

Questions

1. Contributions and Strengths: What is the main impact of the work and its relevance to the audience? Evaluate the methodological soundness, the structural coherence, and highlight the main strengths of the paper.

The paper benchmarks eight configurations, five global Cross-Site and three strictly local, for short-term PV forecasting at newly commissioned plants, under a Leave-One-Plant-Out protocol over 94 Chilean plants from 2014 to 2024, with point and probabilistic metrics stratified by season.

The design is the strongest aspect. The LOPO protocol is genuinely target-blind, zero-shot integrity is enforced by automated tests rather than asserted, and the synthetic cold-start context is a sensible answer to a real problem: how to initialise the input window of a deep sequence model for a plant with no history. Combining a target-blind

evaluation, a contrast against strong local baselines under an identical protocol, and a probabilistic characterisation is not common in this literature, and the paper is explicit about why each is needed.

The reporting is unusually honest. The authors describe the early efficiency-prior bug, with correlation 1.0 against y and a spurious $rRMSE$ near 0.8% that collapsed to about 56%, and they end up rejecting their own hypothesis, since no global model reaches the best local baseline. Reporting a negative result against one's own framing is more useful to the community than the alternative, and I hope it survives into the final version.

The probabilistic section is read carefully. Treating global XGBoost's 0.99 coverage as a defect symmetric to the networks' undercoverage, rather than a virtue, is the correct reading and one that many papers get wrong.

The structure is coherent and the results follow from the protocol. The seasonal stratification produces an operationally useful finding, namely that the uncertainty of a new plant depends strongly on the season of its commissioning. The scale of the corpus and its local relevance also fit the audience well.

2. Weaknesses and Suggestions: Identify fixable areas for improvement (e.g., phrasing, lack of detail) as well as major ones (severe structural or methodological flaws). Provide specific suggestions on what to add, remove, or correct for the final version.

The $rRMSE$ normalisation makes the headline numbers hard to interpret. Values between 62% and 150% invite distrust, and dividing by the window mean with night hours included inflates the index. I would keep the column for continuity and add a capacity-normalised $NRMSE$ beside it, which requires no new experiments.

A temporal point that bears on the headline claim: it is not stated whether the local baselines are trained only on data preceding their evaluation window, or on the plant's full history with the window excised. If the latter, they are fitted partly on the future of what they predict, which would flatter exactly the configuration that wins. Please make this explicit, and if training is not strictly causal, rerun with a rolling origin. The global models have a mirror issue, since LOPO hides the target plant but not the calendar, so the model sees the same hours at neighbouring plants of the same macro-zone. That bias favours the global side and the global side still loses, so it reinforces the conclusion rather than undermining it, but it deserves a sentence.

There is no statistical testing and no dispersion around the medians. The comparison is strictly paired on a shared hourly grid, which is exactly where a paired Wilcoxon signed-rank test applies, or Friedman with Nemenyi for the eight-way ranking. IQRs and per-plant win counts would help. Of everything here, this is what I would prioritise.

Table I is captioned as $n = 33$ plants while Section IV-B implies aggregation over all 94.

If the headline table covers only 33, the abstract's framing around 94 is misleading. I assume a caption error, but it underpins the main result.

A few minor details: the reproducibility section gives no repository URL or DOI, Table II lists seven configurations against the eight of Table I, with local LSTM gone from the seasonal breakdown, and the hyperparameter leakage is disclosed but never bounded, so it would help to know how much it actually moves the numbers.

3. What is the overall score of the submission?

accept